

Priority Queues and Applications

Serob Tigranyan

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1 Introduction

A **Priority Queue** is an abstract data type that operates similarly to a regular queue or stack, but where every element has an associated “priority.” In a priority queue, an element with high priority is served before an element with low priority.

1.1 Implementation: The Binary Heap

In computer science, a priority queue is most efficiently implemented using a **Binary Heap**.

1.1.1 What is a Binary Heap?

A Binary Heap is a complete binary tree that satisfies the **heap property**:

- **Max-Heap:** The value of each node is less than or equal to the value of its parent. The largest element is at the root.
- **Min-Heap:** The value of each node is greater than or equal to the value of its parent. The smallest element is at the root.

1.1.2 Array-Based Implementation

Because a heap is a **complete** binary tree (all levels are filled except possibly the last, which is filled from left to right), it can be stored compactly in a contiguous array without pointers.

For an element at index i :

- **Left Child:** $2i + 1$
- **Right Child:** $2i + 2$
- **Parent:** $(i - 1) / 2$

This layout allows for $O(1)$ access to the top element and $O(\log n)$ time for both insertion and deletion by “bubbling” or “sifting” elements up or down to restore the heap property.

2 C++ Support for Priority Queues

C++ provides built-in support for priority queues through the Standard Template Library (STL).

2.1 The `std::priority_queue` Container

Found in the `<queue>` header, `std::priority_queue` is a container adaptor. It provides $O(\log n)$ insertion and extraction, and $O(1)$ access to the top element.

```
#include <queue>
#include <vector>

// Default: Max-Heap (Highest value at top)
std::priority_queue<int> max_heap;

// Min-Heap (Lowest value at top)
std::priority_queue<int, std::vector<int>, std::greater<int>> min_heap;
```

2.2 Key Properties

- **Underlying Container:** Defaults to `std::vector`, but can use `std::deque`.
- **Ordering:** By default, it uses `std::less<T>`, making it a **Max-Heap**.
- **No Iterator Support:** Unlike `std::vector` or `std::list`, you cannot iterate through a priority queue. You can only access the `top()` element.

3 Technical Challenges: The “Decrease Key” Problem

Standard `std::priority_queue` does **not** support updating the priority of an element already in the queue.

3.1 Solutions:

1. **Lazy Removal:** Instead of updating, push a new entry with the better cost. When popping, check if the node has already been processed with a lower cost; if so, skip it.
2. **`std::make_heap`:** Use a raw `std::vector` with `std::push_heap` and `std::pop_heap`, which allows manual searching or using a custom indexed heap.

4 Example Code: Custom Priority Handling

```
#include <iostream>
#include <queue>
#include <vector>

struct RenderBatch {
    int id;
    int z_index;

    // Min-Heap based on z_index
    bool operator>(const RenderBatch& other) const {
        return z_index > other.z_index;
    }
};

int main() {
    std::priority_queue<RenderBatch,
                        std::vector<RenderBatch>,
                        std::greater<RenderBatch>> draw_queue;

    draw_queue.push({1, 100}); // UI
    draw_queue.push({2, 0});   // Background
    draw_queue.push({3, 50});  // Player

    while (!draw_queue.empty()) {
        RenderBatch batch = draw_queue.top();
        std::cout << "Rendering Batch ID: " << batch.id
                  << " (Z-Index: " << batch.z_index
                  << ")" << std::endl;
        draw_queue.pop();
    }

    return 0;
}
```

5 Applications

As a game engine developer, I personally have used priorities queues in the past so I will demonstrate few of its applications in engine development. Priority queues are vital in game development for managing asynchronous tasks and ensuring deterministic ordering.

5.1 1. Abstracted Render Command Queues

Modern APIs like Vulkan and DirectX 12 have native **CommandQueues**. OpenGL, however, is notoriously single-threaded: its context can only be active on one thread at a time.

- **The Problem:** How do we allow other threads (AI, Physics, Editor) to “call” OpenGL functions?
- **The Solution:** We imitate a Command Queue.
 - **Abstraction:** Create a `GLCommandQueue` class.
 - **Submission:** Worker threads call `gl_submit_command` to push commands (e.g., `DrawMeshCommand`, `UpdateUniformCommand`) to a thread-safe queue.
 - **Execution:** The main thread (Render Thread) calls `gl_execute_commands` in a tight loop, popping and executing the actual GL calls.
- **Role of Priority Queues:** While often a FIFO queue is used for strict ordering, a **Priority Queue** can be used to prioritize “Immediate” commands (like UI or Debug lines) over “Background” commands (like texture mipmap generation).

5.2 2. Multi-threaded Engine Events

In a complex engine or editor, subsystems (Virtual File System, Input, Windowing) need to communicate across threads.

- **Mechanism:** A global `EventManager` where systems register/subscribe to events (e.g., `FileChangedEvent`, `WindowResizeEvent`).
- **The Ordering Challenge:** In a multi-threaded environment, events might be submitted nearly simultaneously from different threads. To maintain determinism, events must be processed in the exact order they occurred.
- **Priority Queue Usage:** By using a **Min-Heap** keyed by a high-resolution **Timestamp**, the engine ensures that a `WindowResizeEvent` that happened at $T = 10ms$ is always processed before a `FileChangedEvent` at $T = 11ms$, regardless of which thread pushed them to the queue first.

5.3 3. 2D Batch Rendering (Z-Index Sorting)

In a 2D engine's batch renderer, draw calls must be ordered correctly to handle layering (parallax, UI, sprites).

- **Mechanism:** Each batch of sprites or geometry is assigned a `z_index`.
- **The Problem:** Different systems (UI, Game World, Particle System) submit draw commands in any order, but they must be rendered from back to front.
- **Priority Queue Usage:** A **Min-Heap** stores batches keyed by their `z_index`.
- **Benefit:** The renderer pops batches in the correct order, ensuring that a “UI” batch with a high Z-Index is always drawn on top of a “Background” batch with a low Z-Index, regardless of submission order.